

Force and Pressure: Motion



LEARNING OUTCOMES

Students will be able to:

- ▶ define motion.
- ▶ identify objects in motion and at rest.
- ▶ describe different types of motion, with examples from daily life.
- ▶ define uniform and non-uniform motion with examples from daily life.
- ▶ define the concept of speed (average speed).
- ▶ calculate the average speed of objects based on data provided.
- ▶ define weight.
- ▶ relate weight of an object with its mass.

Rest and Motion

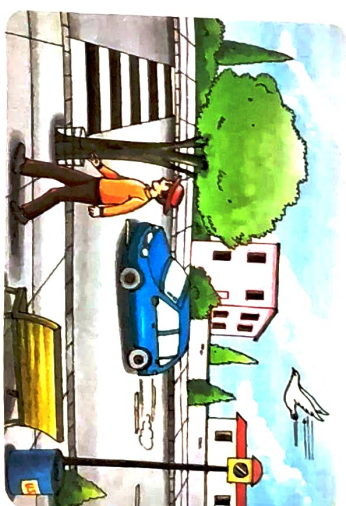
If we look around us, we see many things which are either at rest or moving. For example, houses, trees, electric poles, stone, table, chair, a book lying on a table are said to be at rest. On the other hand, people walking, birds flying, insects crawling, rain drops falling are in motion. Can you imagine an object which is neither at rest nor in motion? No, every object around us is either at rest or in motion.

EXTRA FACT

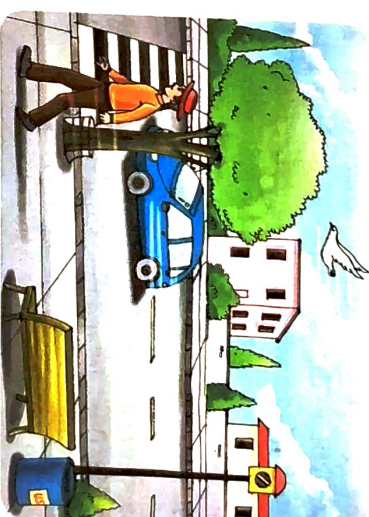
Every object resists changing its state of rest or motion; this is called 'inertia'.

What do we really mean when we say that an object is at rest or in motion? A body is

said to be at rest when its position does not change with time with respect to its surroundings. When the position of a body changes with time with respect to its immediate surroundings, it is said to be in motion. You might have noticed that an electric pole does not change its position with respect to time, whereas a flying bird is not at the same place after some time. So, the electric pole is at rest, whereas the flying bird is in motion.



(a)



(b)

Fig. 2.1 Objects at rest and in motion

Rest and Motion are Relative Terms

Whenever we speak of a body at rest or in motion, we always say this in comparison to a second body, which is known as the **reference body**. For example, when we say that a car is at rest or in motion, we describe the state of the car with respect to the Earth—the reference body in this case is the car and we consider the Earth to be at rest. But, we know that the Earth spins around its own axis once in 24 hours and also moves around the sun. How can the buildings, trees, roads, towers, etc. situated on the moving Earth remain at rest? They

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are at rest because they never change their positions with respect to their surroundings. Let us take another example. Suppose, you are travelling by a train. If you compare your position in relation to the other passengers sitting beside you or the walls and roof of the train, your position is not changing. In other words, you are at rest in relation to these objects. But if you compare your position in relation to a person standing on the platform or trees and animals in the field, you are in motion.



Fig. 2.2 Relative motion

KNOW YOUR SCIENTIST

Newton
(1643–1727)



Sir Isaac Newton, an English physicist and mathematician, is famous for his laws of motion and the 'apple falling down' episode. Newton formulated the basic laws of gravitation which are now considered universal. He published a book titled "Principia Mathematica" in 1687 in which he summarised many of his original ideas in mathematics and physics, including the laws of motion and the theory of gravitation.

Thus, we can say that a body can be in motion for one set of objects, and at the same time, it can be at rest for another set of objects. Therefore, rest and motion are considered to be relative terms.

Scalar and Vector Quantities

In physics, we come across innumerable quantities of different kinds, such as length, mass, time, speed, velocity, acceleration, etc. These are called physical quantities. Most of the physical quantities are divided into two types:

- Scalar quantities
- Vector quantities

A physical quantity that can be described only by its magnitude is called a **scalar quantity**. Mass, length, time, volume, distance, energy, speed, etc. are examples of scalar quantities.

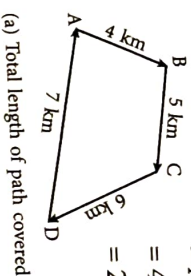
A physical quantity that needs both magnitude and direction for its complete description is called a **vector quantity**. Displacement, velocity, acceleration, force, weight, etc. are examples of vector quantities.

Distance and Displacement

The actual length of the path covered by a moving body in a certain interval of time is called the distance travelled by the body. Since it is just the magnitude of distance, taken in any direction, it is considered to be a scalar quantity. Its SI unit is metre (m).

Suppose, a body covers a distance of 4 km while moving from A to B, 5 km from B to C, 6 km from C to D and 7 km from D to A. The total length of the path covered by the body—

$$= AB + BC + CD + DA \\ = 4 + 5 + 6 + 7 \\ = 22 \text{ km}$$



The shortest distance covered in a particular direction by a body while moving from one point to another is called its **displacement**. It is a vector quantity since it has both magnitude and direction. The SI unit of displacement is metre (m).

Suppose, a body travels from a point A to B (3 km) due east and then from B to C (4 km) due north.

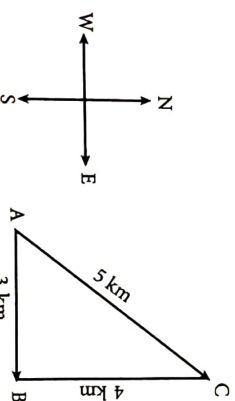


Fig. 2.3 Distance and displacement

The distance travelled by the body is $AB + BC = 3 + 4 = 7 \text{ km}$

Whereas the displacement of the body is—

$$AC = \sqrt{AB^2 + BC^2} \\ = \sqrt{3^2 + 4^2} \\ = 5 \text{ km due north-east.}$$

Now, if the body returns to point A from C due south-west, the displacement of the body will be zero, whereas the distance travelled will be $AB + BC + CA = (3 + 4 + 5) = 12 \text{ km}$.

Table 2.1 Difference between Distance and Displacement

Distance	Displacement
1. The actual length of the path covered by a moving body in a certain interval of time is called the distance travelled by the body.	1. The shortest distance covered by a body in a particular direction while moving from one point to another is called its displacement.
2. It is a scalar quantity.	2. It is a vector quantity.
3. Distance is always positive.	3. Displacement can be both positive and negative.
4. Distance travelled by a moving body in a certain interval of time can never be zero.	4. Displacement of a moving body in a certain interval of time can be zero.
5. Distance $\geq$ Displacement	5. Displacement $\leq$ Distance

Speed and Velocity

The speed of a moving body is defined as the distance travelled by the body per unit time.

$$\text{That is, Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

Speed is a scalar quantity. The SI unit of speed is m/s.

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EXTRA FACT

In a vehicle, speed is measured by a speedometer while distance is measured by an odometer.

Average Speed

Average speed of a body is defined as the ratio of the total distance travelled to the total time taken by the body.

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

Its SI unit is m/s.

The velocity of a moving body is the rate of change of its position with time in a particular direction. In other words, the displacement of a moving body per unit time is called the velocity of the body.

$$\text{Hence, Velocity} = \frac{\text{Displacement}}{\text{Time taken}}$$

Velocity is a vector quantity. Its SI unit is metre per second (m/s). Being a vector quantity, velocity can be negative. Take an example of two cars moving with a speed of 40 km/h, one towards the east and other towards the west. Then velocity of one car with respect to the other is negative.

EXTRA FACT

Space rockets are some of the fastest moving vehicles which reach speeds of around 28,000 km/h as they orbit the earth.

Table 2.2 Difference between Speed and Velocity

Speed	Velocity
1. Distance travelled by a body per unit time is called speed.	1. Distance travelled by a body per unit time in a given direction is called velocity.
2. It is a scalar quantity.	2. It is a vector quantity.
3. Speed can never be negative.	3. Velocity can be negative.

Motion Along a Straight Line

The motion along a straight line can be divided into two types.

Uniform Motion

When a body covers equal distances in equal intervals of time along a straight line then the motion of the body is said to be uniform. For example, if a car travels 80 km every hour or 40 km in every half an hour in a particular direction, then the motion of the car is said to be uniform.

Non-uniform Motion

When a body covers unequal distances in equal intervals of time along a straight line then the motion of the body is termed as non-uniform. For example, when a ball is rolled on the ground or when the brakes are applied on a fast moving car, it describes a motion that is non-uniform. It is because the distances covered by the ball or the car in equal intervals of time are not the same.

Types of Motion

Different objects perform different kinds of motion. For example, a falling stone moves along a straight line, an electric fan rotates

round in a circular motion. We shall classify the motion of different objects into the following categories:

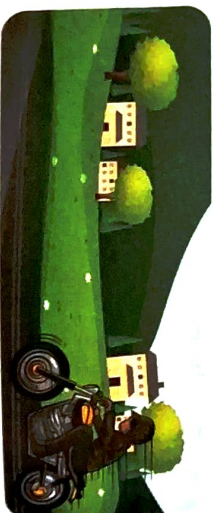
1. Translatory Motion

The motion in which every point on the moving body moves through the same distance in the same interval of time is called translatory motion. Translatory motion can be of two types:

- Rectilinear motion
- Curvilinear motion.

a. Rectilinear Motion

If the motion of a body is along a straight line, its motion is called rectilinear motion. For example, a vehicle moving on a straight road, a bird flying along a straight line, a child sliding down a rope, a falling stone, firing a bullet from a gun, pulling out a drawer, marching of soldiers in a parade, runners in a 100-metre track, etc. are in rectilinear motion. In all these examples, we see that the objects change their position with time in a straight line.



(a) A vehicle moving on a straight road

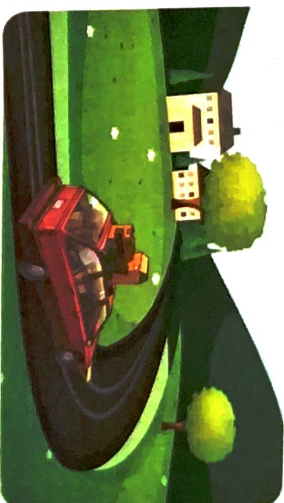


(b) Firing a bullet from a gun

Fig. 2.4 Rectilinear motion

b. Curvilinear Motion

If the motion of a body is along a curved path, its motion is called curvilinear motion. A basket ball being thrown into a basket, a car taking a turn on a road, a train moving along a curved track, a javelin or discus being thrown by an athlete are examples of curvilinear motion.



(a) A car taking a turn on a road



(b) A basket ball being thrown into a basket

Fig. 2.5 Curvilinear motion

2. Rotatory Motion or Circular Motion

When a body moves about a fixed axis (or a fixed point) without changing its position, the body is said to have a rotatory motion. During the motion, the distance of every point on the moving body from the fixed axis (or the fixed point) remains the same. But in the same interval of time, every

point on the rotating body moves through a different distance. For example, a spinning wheel (*charakha*), a spinning top, a potter's wheel, the motion of the blades of a fan, the motion of the second's hand in a clock, a child on the merry-go-round produce rotatory motion.

EXTRA FACT

When a roller coaster travels through a loop, a special kind of force called 'centripetal force' keeps the people pinned to their seats.



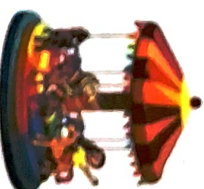
(a) Spinning wheel or charakha



(b) A spinning top



(c) Motion of a potter's wheel



(d) Motion of a merry-go-round

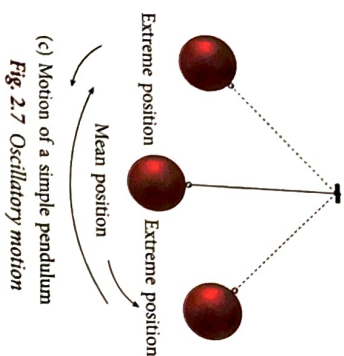
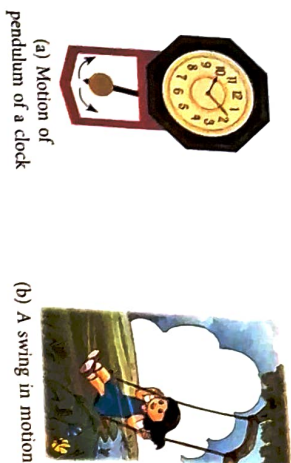
Fig. 2.6 Rotatory motion

3. Oscillatory Motion

A motion in which a body moves 'to and fro' or 'back and forth' about its mean position (rest position) is called oscillatory motion. For example, motion of a simple pendulum, motion of a swing, motion of the pendulum of a wall clock, motion of the piston of a motor car, etc. represent oscillatory motion.

EXTRA FACT

Naturally oscillating objects slow down and stop because of friction.



4. Vibratory Motion

A motion in which a part of a body moves to and fro in a definite style while the rest of the body always remains fixed is called vibratory motion.

It is important to note the difference between an oscillation and a vibration. In an oscillation, the whole body moves to and fro along a definite path. In vibration, on the other hand, a certain part or parts of the body always remain fixed and do not move while the rest of the body moves to and fro in a definite style. For example, when we pluck the strings of a guitar, only the string moves, whereas the guitar remains stationary. In fact, all musical instruments, such as the sitar, violin, tabla, drum, etc. produce vibratory motion.

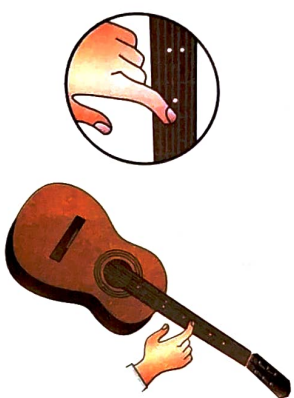
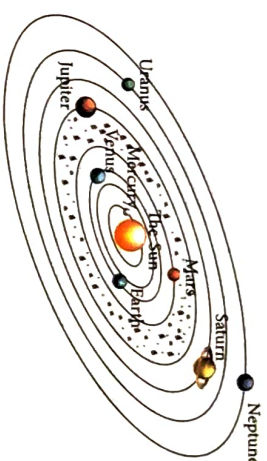


Fig. 2.8 Motion of the strings of a guitar is vibratory motion

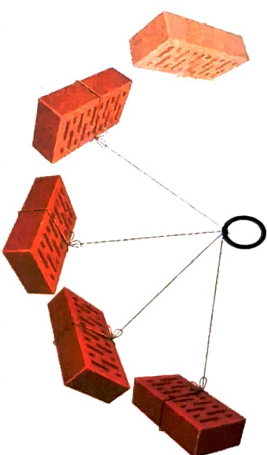
5. Periodic Motion

Any motion that repeats itself at regular intervals of time is known as periodic motion. For example, motion of a pendulum, motion of a planet around the sun, motion of a mass suspended by a string or spring, motion of the strings in the stringed instruments, the motion of a swing, etc. are periodic.

It should be noted that all periodic motions need not be oscillatory but all oscillatory motions are periodic.



(a) Motion of planets around the sun



(b) Motion of a mass suspended by a string

Fig. 2.9 Periodic motion

6. Non-periodic Motion

A motion which does not repeat itself after a fixed interval of time is called non-periodic motion.

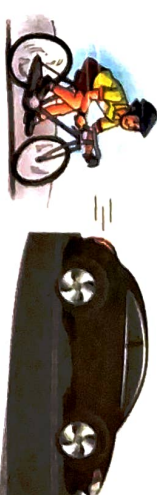
Motion of our lungs during breathing, motion of a rolling cricket ball on the ground, swinging of our arms or legs as we walk, motion of the wheels of a moving car when brakes are applied are repetitive, but not periodic.



Fig. 2.10 Non-periodic motion

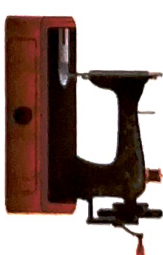
7. Complex or Multiple Motion

A motion which is a combination of two or more types of motion is called multiple motion.



(a) A boy riding a bicycle

(b) Wheels of a moving vehicle



(c) Movements of the parts of a sewing machine
Fig. 2.11 Multiple motion

When you ride a bicycle, the wheels of the bicycle execute both rotatory and translatory motions simultaneously. So, it is a multiple motion. Wheels of all vehicles execute both rotatory and translatory motions simultaneously. A sewing machine also performs multiple motion.

The motion in which there is simultaneous occurrence of both rotational and translational motions is called rolling motion.

A few other examples of multiple motion are:

- Motion of the earth.
- Motion of an electric drill.
- A ball rolling on the ground, etc.

EXPERIMENT-1

Producing oscillatory motion

Take a rubber band. Tie a stone at one end of the rubber band and fix the other end to a rigid support like a door knob or a handle. Now stretch the rubber band by pulling down the stone by a certain length and release it. Observe the motion of the stone.

You will find that the stone is executing to and fro motion, i.e., oscillatory motion.

8. Random Motion (Zigzag or Irregular Motion)

A motion in which a moving object frequently changes its direction of motion is called random, zigzag or irregular motion.

The motion of a mosquito in flight, the motion of smoke particles, etc. are examples of random motion. The most common example of this type of motion is Brownian motion. It is the zigzag motion of small pollen grains suspended in water, first observed by Robert Brown.

KNOW YOUR SCIENTIST

Robert Brown
(1773–1858)



The scientist, Robert Brown, noted 'random motion of tiny particles suspended in a liquid or gas' and reported this movement for the first time. So this movement is called 'Brownian Movement.'

The Motion of a Simple Pendulum as Periodic Motion

If a heavy point mass is suspended by a weightless, inextensible and perfectly flexible string from a rigid support, then this arrangement is called a simple pendulum. In practice, however, these requirements cannot be fulfilled.

A practical pendulum consists of a small metallic solid sphere suspended by a fine silk thread from a rigid support. This is the

practical simple pendulum, which is nearest to the ideal simple pendulum. The metallic sphere is called the 'bob'. When the bob is displaced slightly to one side from its mean position and released, it oscillates about its mean position in a vertical plane.

EXTRA FACT

Isn't it amazing that pendulums are used to protect newly-built skyscrapers from damage by earthquakes? Huge pendulums installed at the upper floors work by automatically swaying in the opposite direction to the motion of the quakes, thus stabilizing the buildings.

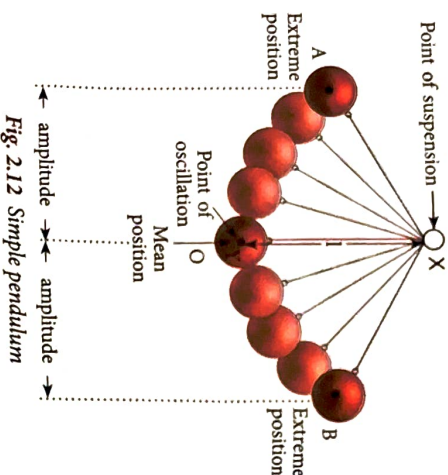


Fig. 2.12 Simple pendulum

Terms Related to a Simple Pendulum

Different terms related to the simple pendulum are discussed below.

- Point of suspension:** It is the fixed point (X) in a rigid support from which the pendulum is suspended.
- Point of oscillation:** The centre of gravity of the pendulum (Y) is called the point of oscillation.

As the thin silk thread is very light, the centre of the bob becomes the centre of oscillation.

- Length of the pendulum:** This is the distance between the point of suspension and the point of oscillation of the pendulum (XY).
- Mean position (or rest position):** It is the position of the bob when the freely suspended pendulum is at rest. O is the mean position or rest position of the pendulum.

v.

Extreme position (or turning point): The positions of the bob at the maximum distance on both sides of the mean position up to which the pendulum can oscillate are called extreme positions. A and B are the extreme positions on either side of the pendulum.

vi.

One oscillation: One complete to and fro movement of a pendulum about its mean position is known as one oscillation or vibration. The movement from O to A, then from A to B and back to O constitutes one oscillation.

vii.

Amplitude: During an oscillation the maximum displacement of the pendulum from its mean position is called its amplitude. OA and OB are the amplitudes of the simple pendulum.

viii.

Time period: Time taken by a pendulum to complete one oscillation is called its time period. It is denoted by T.

ix.

Frequency: The number of vibrations or oscillations made by a pendulum in one second is called the frequency of the pendulum. It is denoted by 'f' and its unit is hertz (Hz).

$$\therefore f = \frac{1}{T}$$

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EXPERIMENT-2

Finding the time period of a simple pendulum

Aim: To determine the time period of a simple pendulum.

Apparatus: A metal bob, silk thread, a stand and clamp arrangement, a stopwatch.

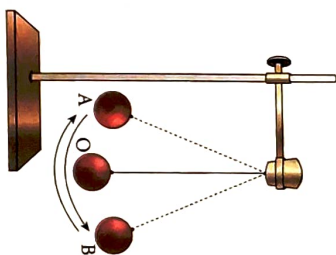


Fig. 2.13 Simple pendulum

Procedure:

- Set up a simple pendulum as shown in the figure.
- Set the simple pendulum in motion.
- Start the stopwatch when the bob is at one of the extreme positions.
- When the bob comes back to this position, one oscillation is complete.
- As the bob completes 20 such oscillations, stop the stopwatch and note the time 't' for 20 oscillations.

Observation:

Time taken for 20 oscillations = t

$$\text{Inference: Time period } T = \frac{t}{20} \text{ s}$$

Force

A physical cause which changes either the size or shape, or sets a body into motion or brings a body to rest is called force.

Examples:

- If an iron nail is hammered, it changes in shape.

ii. If a ball is pushed, it starts rolling, that is, the ball starts to roll by a push.

iii. The driver of a vehicle applies brakes to stop his vehicle.

Units: SI Units : newton (N)

CGS unit: dyne

$1 \text{ N} = 10^5 \text{ dyne}$

Other units : kgf, gf

Force can be applied on a body in two different ways:

i. By touching the body (contact force).

Examples: push, pull, squeeze, stretch, friction

ii. Without touching, that is, from a distance (non-contact force).

Examples: Attraction or repulsion between bodies by magnets and electric charge; gravitation force.

Force is a vector quantity.

EXTRA FACT

When you push a wall, a force of equal magnitude but in opposite direction is applied by the wall on you! But you cannot move the wall because the force applied by the rest of the building on the wall is greater and keeps it in the same position.

Weight

This physical quantity is also a kind of force. In everyday life, we use the term weight instead of mass, but in physics, mass and weight have different meanings.

The force with which the earth attracts a body is called its weight. The SI unit

of weight is newton (N). Weight is also measured in kilogram-force (kgf). It is the force with which one kilogram mass is attracted by the earth.

It has been found that $1 \text{ kgf} = 10 \text{ N}$

The weight of a body is equal to the product of its mass and the acceleration due to gravity.

$$W = mg$$

Where 'm' is the mass of the object and 'g' is the acceleration due to gravity.

Let us perform the following experiment to understand the meaning of weight.

EXPERIMENT-3

The force of gravity

Hold a small stone in your hand. When it is released, it falls vertically downwards and hits the ground with a certain force and its speed becomes zero. This suggests that an object changes its speed or position only when a force is applied on it.

Which force is applied in case of the stone?

It is the force exerted by the earth towards its centre. It is called the **force of gravity**. It is also called the weight of an object. The acceleration (change in velocity per unit time) of the stone is called **acceleration due to gravity** which is usually denoted by the letter 'g'.

Variation in Weight of a Body

The weight of a body is not the same at all places on the earth. It is maximum at the poles and minimum at the equator because the gravitational pull of the Earth

is maximum at the poles and minimum at the equator, i.e., the value of acceleration due to gravity is maximum at the poles and minimum at the equator. Thus, we see that the weight of a body depends on the value of 'g'.

It is found that the force of gravity decreases with altitude. The weight of a body decreases as we go higher and higher because the value of acceleration due to gravity decreases as the distance from the centre of the earth increases.

Instruments Used to Measure Weight

The instrument commonly used to measure weight is a spring balance. Spring balance is a kind of balance having a helical spring.

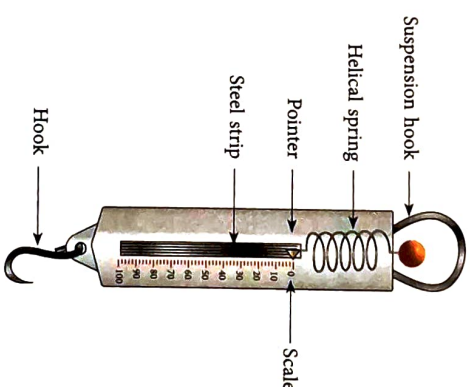


Fig. 2.14 Spring balance

Electronic Balance

It is a very sensitive and accurate balance. The object to be measured is placed on the

platform and the weight can be read on the screen. As this balance is very sensitive and accurate, it is used in laboratories and in jewellery and grocery shops.

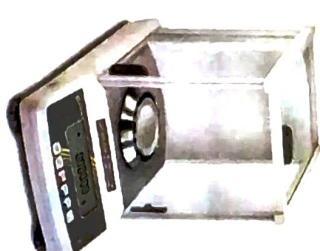


Fig. 2.15 Electronic balance

From our previous discussion, we know that mass is the amount of matter contained in a body. It is a constant quantity for a body irrespective of its position and surroundings.

The mass of a body can be expressed in different units like kilogram (kg), gram (g), quintal, tonne, etc. but the SI unit of mass is kg.

Table 2.3 Difference between Mass and Weight

Mass	Weight
1. Amount of matter in a substance.	1. The force of gravity acting on a body.
2. SI unit of mass is kilogram (kg).	2. SI unit of weight is Newton (N).
3. Mass remains constant at a place.	3. Weight varies from place to place.
4. Mass is measured by a physical or beam balance.	4. Weight is measured by using a spring balance.

Solved Numericals

Let Us Summarise

Let Us Summarise

Example 1: A train moving with uniform speed covers a distance of 120 m in 2 s. Calculate—
i. the speed of the train,
ii. the time it will take to cover 240 m.

Solution:

$$\text{Speed of the train} = \frac{\text{Distance travelled}}{\text{Time taken}} = \frac{120}{2} = 60 \text{ m/s}$$

$$\text{Time taken to cover a distance of 240 m} = \frac{\text{Distance}}{\text{Speed}} = \frac{240}{60} = 4 \text{ s}$$

Example 2: A train takes 3 hours to travel from Agra to Delhi with a uniform speed of 70 km/h. Find the distance between the two cities.

Solution: Distance = Speed of the train $\times$ Time taken = $70 \times 3 = 210 \text{ km}$

Example 3: A train takes 2 hours to reach from station A to station B and then 3 hours to return. The distance between the two stations is 200 km. Calculate the average speed of the train.

Solution:

$$\text{Average speed of the train} = \frac{\text{Total distance travelled}}{\text{Total time taken}} = \frac{(200 + 200)}{(2 + 3)} = \frac{400}{5} = 80 \text{ km/h}$$

Example 4: Rajdhani Express moves at a speed of 120 km/h. How long will it take to cover a distance of 15 km?

Solution: Distance covered = 15 km

Speed = 120 km/h

$$\text{Time} = \frac{\text{Distance covered}}{\text{Speed}} = \frac{15}{120} = \frac{1}{8} \text{ h} = \frac{1}{8} \times 60 = 7.5 \text{ min} = 7 \text{ min } 30 \text{ s}$$

Example 5: A motorcycle is running at a constant speed of 15 m/s.

i. How much distance will it cover in 45 s?

ii. How much time will it take to cover 36 km?

Solution:

i. Speed of motorcycle = 15 m/s

Time = 45 s

$$\therefore \text{Distance covered in 45 s} = \text{Speed} \times \text{Time} = 15 \times 45 = 675 \text{ m}$$

ii. Speed = 15 m/s = $15 \div \frac{5}{18} \text{ km/h} = 54 \text{ km/h}$

Distance = 36 km

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{36}{54} = \frac{2}{3} \text{ h} = 40 \text{ min}$$

Example 6: Calculate the weight of a body having mass 15 kg. (Take $g = 10 \text{ m/s}^2$)

Solution: Mass (m) = 15 kg

$g = 10 \text{ m/s}^2$

$$\text{Weight} = mg = 15 \text{ kg} \times 10 \text{ m/s}^2 = 150 \text{ N}$$

1. The physical quantity that can be described completely only by its magnitude is called a scalar quantity, while the physical quantity that needs both magnitude and direction for its complete description is called a vector quantity.

2. The actual length of the path travelled by a body in a certain interval of time is called the distance. It is a scalar quantity.

3. The shortest distance from the initial to the final position of the body is called its displacement. It is a vector quantity.

4. The speed of a moving body is the distance travelled by it per unit time. It is a scalar quantity.

5. The rate of change of displacement of a body with time is its velocity. It is a vector quantity.

6. The motion in which every point on the moving body moves through the same distance in the same interval of time is called translatory motion.

7. When a body moves about a fixed point or axis without changing its position, the body is said to have a rotatory motion.

8. A motion in which a body as a whole moves to and fro about its mean position is called oscillatory motion.

9. The motion in which a part of a body moves to and fro in a definite style, while the rest of the body remains at rest is called vibratory motion.

10. A motion that repeats itself at regular intervals of time is known as periodic motion.

11. A motion which is a combination of two or more types of motion is called multiple motion.

12. The motion in which there is simultaneous occurrence of both rotational and translational motion is called rolling motion.

13. The motion in which the moving object frequently changes its direction of motion is called random motion.

14. If a heavy point mass is suspended by a weightless, inextensible and perfectly flexible string from a rigid support, then this arrangement is called a simple pendulum.

15. The time taken by a pendulum to complete one oscillation is called its time period.

16. Force is that physical cause that tends to change the shape or size of a body or sets a body into motion or brings a body to rest.

17. There are two types of force—contact force and non-contact force.

18. Weight of a body is that force which pulls the body towards the Earth. The SI unit of weight is newton (N).

19. Weight of a body at a given place is measured by a spring balance.

Objective Evaluation

REVIEW QUESTIONS

A. Choose the correct answer:

1. The wall of your classroom is in a state of—
a. motion ☒ b. rest c. neither rest nor motion d. None of these
2. Translatory motion is of—
a. one type ☒ b. two types c. three types d. four types
3. The motion of a pendulum is an example of—
a. translatory motion b. rotatory motion
☒ c. oscillatory motion d. curvilinear motion
4. Velocity is a—
a. scalar quantity ☒ b. vector quantity c. normal quantity d. None of these
5. Distance is a—
☒ a. scalar quantity b. vector quantity c. normal quantity d. None of these
6. A moving vehicle on a straight road is an example of—
a. rotatory motion b. vibratory motion c. periodic motion ☒ d. rectilinear motion
7. The motion of a dust particle in air is an example of—
a. rectilinear motion b. curvilinear motion
c. oscillatory motion ☒ d. random motion
8. Force is a physical cause that—
a. changes the shape of a body b. brings a body to rest
c. sets a body into motion ☒ d. All of these

9. Weight of a body—

- a. remains the same everywhere on the Earth
- b. will be the same on the moon
- c. is the pressure exerted by it
- d. is measured by a spring balance

B. Fill in the blanks:

1. A book lying on a table is in a state of rest.
2. All oscillatory motions are periodic.
3. The value of acceleration due to gravity is 9.83 m/s² at the poles.
4. An apple falling from a tree is an example of random motion.
5. Displacement is a vector quantity.
6. When the Earth completes one revolution, the displacement of the Earth is 0.
7. The time taken by the bob of a pendulum to complete one oscillation is called its frequency.
8. A potter's wheel has circular motion.
9. Force is a vector quantity.
10. Weight of a body becomes less on top of high mountain.

C. Write True (T) or False (F) against the following statements:

1. A spinning top has a rectilinear motion. ☒ F
2. When a body covers equal distances in equal intervals of time in the same straight line, the body is said to be in a uniform motion. ☒ T
3. The rate of change of displacement with time is called speed. ☒ F
4. Displacement is a scalar quantity. ☒ F
5. Speed can be positive, zero or negative. ☒ T
6. Rotation of the Earth is a periodic motion. ☒ T
7. Fruits falling from a tree is an example of random motion. ☒ T
8. Weight of a body remains the same if the mass of the body increases. ☒ F
9. Force of friction between two bodies increases the speed of a moving body. ☒ F

D. Match the following:

Column A

1. A bouncing ball
2. A spinning top
3. A swing in motion
4. A plucked string of a musical instrument
5. A ball rolling on the ground
6. Velocity
7. Distance
8. Speed
9. Time
10. Force

Column B

- a. Second
- b. Vector
- c. Rectilinear motion
- d. Multiple motion
- e. Vibratory motion
- f. Oscillatory motion
- g. Rotatory motion
- h. Metre
- i. Distance ÷ Time
- j. Metre per second

E. With proper reasoning, choose the odd one out:

1. bouncing ball, spinning top, oscillating pendulum, study table.
2. displacement, velocity, speed, average speed.
3. a falling stone, a child sliding down a slope, firing a bullet from a gun, a girl swinging on a swing.
4. motion of an electric drill, a rolling ball, wheels of a moving car, oscillating pendulum of a wall clock.

Subjective Evaluation

F. Answer the following questions:

1. What do you mean by a body at rest?
2. What do you mean by translatory motion? Give one example.
3. What is meant by circular motion? Give an example.
4. Why are rest and motion called relative terms?
5. What is the difference between rectilinear motion and curvilinear motion?
6. 'An oscillatory motion is always periodic, but a periodic motion is not always oscillatory'. Explain the statement with an example.

7. State the body which is at rest and the body which is in motion in the following situations—

- i. You are travelling in a train.
 - ii. The revolution of the Earth around the sun.
8. Name the type/types of motion in the following:

- i. A car moving on a straight road
- ii. A basketball thrown in a basket
- iii. A girl skipping a rope
- iv. Earth moving around the sun
- v. Smoke particles moving upwards from a chimney
- vi. A swing in motion
- vii. Blades of a moving electric fan
- viii. The wheels of a moving vehicle
- ix. Spinning *charkha*
- x. Motion of the piston of a car
- xi. Motion of a potter's wheel
- xii. A flying mosquito

G. Define the following:

1. Translatory motion
2. Force
3. Oscillatory motion
4. Weight of a body
5. Periodic motion
6. Displacement
7. Velocity
8. Time period of a pendulum

H. Differentiate between:

1. Rest and motion
2. Rectilinear motion and curvilinear motion
3. Uniform and non-uniform motion
4. Mass and weight
5. Speed and velocity
6. Distance and displacement
7. Force and weight
8. Periodic and non-periodic motion

I. Give reasons for the following:

1. Motion of the wheels of a moving car is a multiple motion.
2. Revolution of the moon around the Earth is a periodic motion.
3. A particle rotating in a circle may have constant speed but its velocity changes continuously.
Velocity is a vector quantity and both magnitude and direction for a particle rotating continuously in a circle with constant speed changes.
4. We weigh more at sea level than on top of a high mountain.

Numericals

1. A car travels from town A to town B that is 150 km away in 3 hours and from town B to town C which is 120 km away in 2 hours. Find the average speed of the car.
[Ans: 54 km/h]
2. Calculate the distance travelled by a man walking at a speed of 5 km/h in 18 minutes.
[Ans: 1.5 km]
3. A bicycle moves at a speed of 18 km/h. How much time will it take to cover a distance of 3 km?
[Ans: 10 minutes]
4. Calculate the weight of a body of mass 13 kg and $g = 10 \text{ m/s}^2$.
[Ans: 130 N]
5. The value of acceleration due to gravity on the moon is approximately 1.60 m/s^2 . A body weighs 320 N on the moon, what is the mass of the body?
(Ans: 200 kg)

Life Skills

While watching a cricket match how many kinds of motion of the ball can you identify? the batsman has struck it?